1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: In localized prostate cancer treated with curative intent, cribriform architectural fraction (CAF) interacts with CCP score to determine progression-free interval (PFI). Specifically, CAF-high tumors with high CCP scores will have the shortest PFI, while CAF-high/CCP-low tumors will have intermediate PFI, and CAF-low tumors (regardless of CCP) will have the longest PFI. Mechanistically, CAF reflects architecture-associated proliferative programs; when these programs are amplified (high CCP), the proliferative drive accelerates progression. Testable predictions: (1) A statistically significant CAF*CCP interaction term in a Cox model predicting PFI after adjusting for Gleason grade, preoperative PSA, PTEN loss, and ERG status. (2) Stratified analyses show CAF-high/CCP-high has the worst PFI, followed by CAF-high/CCP-low, with CAF-low groups faring best. (3) The interaction remains significant after adjusting for known clinical covariates; (4) PTEN loss or ERG status may modulate the interaction magnitude as secondary modifiers. Clinically, this supports risk-stratified approaches for adjuvant therapy in CAF-rich tumors with high proliferative signals.

Element	Value
Imaging feature	Cribriform architectural fraction (CAF)
Anchor gene(s)	CCP score
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CCP score: CCP score (molecular proliferation) — I built on the idea that architecture and proliferation are complementary and that architecture may potentiate proliferative programs.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Gleason grade	Supporting modifier from knowledge graph
Secondary	Preoperative PSA	Supporting modifier from knowledge graph

Secondary	PTEN loss	Supporting modifier from knowledge graph
Secondary	ERG fusion	Supporting modifier from knowledge graph
Secondary	Decipher score	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Gleason grade: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Preoperative PSA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.